

Patrick Kariuki

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EDUCATION

Johns Hopkins University

Bachelor of Science in Computer Science

Baltimore, MD

Expected May 2026

- **Relevant Coursework:** Data Structures, Algorithms, Full-Stack JavaScript, Artificial Intelligence, Computer Systems, Object-Oriented Software Engineering, Databases, Web Security, Operating Systems, Deep Learning.

EXPERIENCE

Course Assistant

Johns Hopkins University

August 2024 – Present

Baltimore, MD

- Mentor over 100 students across multiple computer science courses including Data Structures (Fall 2025 – Spring 2026), AI-Enabled Software Engineering (Spring 2025), and Intermediate Programming in C/C++ (Fall 2024), conducting office hours and providing technical guidance.
- Guided 6-person project teams as technical PM for AI-Enabled Software Engineering course, coordinating milestone delivery, providing feedback on generative AI application development, LLM integration, and deployment architecture.

Tech Developer Intern

Sponsors for Educational Opportunity (SEO)

May 2024 – August 2024

New York, NY

- Built StarSight, a full-stack stargazing prediction platform integrating real-time weather data, light pollution metrics, and astronomical databases to provide location-based visibility forecasts for 50 beta users across 3 states.
- Engineered Flask REST API with secure user authentication and PostgreSQL database; implemented caching layer reducing API response time from 800ms to 200ms for repeated location queries.
- Trained constellation recognition model using Principal Component Analysis on 10,000 labeled astronomical images, achieving 85% accuracy; deployed via Docker with Ansible-managed CI/CD pipeline.

PROJECTS

SDB | C++, Docker, Linux

March 2026 – Present

- Building x86-64 debugger with breakpoint support, stack unwinding, and source-level stepping by implementing ptrace system calls and parsing debug information.

2D Game Engine | C++, Raylib, OpenGL, Lua

January 2026 – Present

- Designed component-based game engine featuring entity-component-system design, collision detection and resolution, render layers, sprite system, animation framework, and custom memory allocators.

GoCI | Go, Git

January 2026 – February 2026

- Programmed CLI continuous integration tool for Go projects with automated build, test, format checking, and git push workflows; actively used for personal project deployment automation.

LeetScribe | React, Chrome Extension API, Convex, GitHub API

June 2025 – October 2025

- Developed Chrome extension that automatically captures LeetCode solutions and generates structured documentation using AI, with GitHub API integration for version-controlled interview preparation notes.

LEADERSHIP

President, Software Engineering Club

May 2025 – Present

- Secured \$500 in corporate sponsorship and organized 5+ technical workshops on career prep, interview skills, and AI-enabled software engineering for 75+ student members.
- Founded HopBuilds, a project-based learning initiative where students build real applications for academic credit; led development of Hopkins Lost and Found platform deployed campus-wide.
- Grew active membership 40% through hands-on coding events and industry networking sessions.

TECHNICAL SKILLS

Programming: Python, C/C++, Java, JavaScript/TypeScript, Kotlin, Swift, Go, Lua, HTML/CSS

Frameworks: React, Node.js, Flask, Django, FastAPI

Databases: PostgreSQL, MongoDB, Firebase, SQLite

Developer Tools: Git, Docker, Kubernetes, Ansible, Linux